Evidence Preservation and Analysis

Introduction

The objective of this task was to simulate digital forensic evidence collection and preservation in a Security Operations Center (SOC) environment. The exercise involved collecting volatile system data from a Windows virtual machine using **Velociraptor**, preserving memory evidence, and verifying the integrity of the collected data through cryptographic hashing.

Maintaining evidence integrity is critical in incident response because it ensures that forensic artifacts remain unchanged and admissible for investigation or legal purposes.

Environment Setup

The investigation environment consisted of the following components:

Component	Description
Investigation Tool	Velociraptor
Forensic Tool	FTK Imager
Target System	Windows Virtual Machine
Investigator Role	SOC Analyst

The SOC analyst used Velociraptor to perform remote forensic artifact collection from the Windows endpoint.

Volatile Data Collection using Velociraptor

Step 1: Accessing Velociraptor Dashboard

The Velociraptor web interface was accessed through the browser.

Example access URL:

```
https://127.0.0.1:8889
```

After authentication, the **Hunts** section of Velociraptor was opened to run forensic artifacts on the client system.

The investigation focused on collecting **active network connections**, which represent volatile data that may disappear once the system is shut down.

Step 2: Running Network Artifact Collection

The following Velociraptor artifact was executed:

```
Windows.Network.Netstat
```

This artifact collects network connection information similar to the `netstat` command on Windows systems.

The artifact queries the endpoint for:

- Active network connections
- Listening ports
- Process identifiers associated with connections

The collected data was exported as a **CSV file** for forensic analysis.

Memory Evidence Collection

Step 3: Performing Memory Acquisition

To preserve volatile system memory, the following artifact was executed in Velociraptor:

```
Artifact.Windows.Memory.Acquisition
```

This artifact collects a full memory dump from the target system.

Memory acquisition is important because RAM may contain:

- Running processes
- Malware artifacts
- Encryption keys
- Network session information
- Suspicious injected code

The memory image was successfully collected and stored by Velociraptor.

Evidence Export

After the hunt finished executing, Velociraptor generated downloadable results.

The collected artifacts included:

- Network connection logs
- Memory acquisition results
- Container files storing forensic artifacts

These files were exported for forensic analysis.

Evidence Integrity Verification

After collecting the memory artifact, a **cryptographic hash** was generated to verify evidence integrity.

Hashing ensures that the collected forensic evidence has not been altered.

The following command was executed in the system terminal:

```
certutil -hashfile "All Windows.Memory.Acquisition.csv" SHA256
```

The command generated a SHA256 hash value for the memory artifact.

Generated hash:

```
8d0ab37cb71ee6121385ca54a9d2e3d53b23afa4b95816a8da154d481d79c9
2f
```

This hash uniquely identifies the collected evidence and can be used later to verify that the file has not been modified.

Chain of Custody Documentation

Proper documentation is essential when handling digital evidence. A chain-of-custody record was created to document the collection details.

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-X Dump	SOC Analyst	2026-03-16	8d0ab37cb71ee6121385ca54a9d2e3d53b23afa4b95816a8da154d481d79c92f

This record ensures accountability and traceability for forensic evidence.

Evidence Preservation

To maintain evidence integrity, the following preservation procedures were applied:

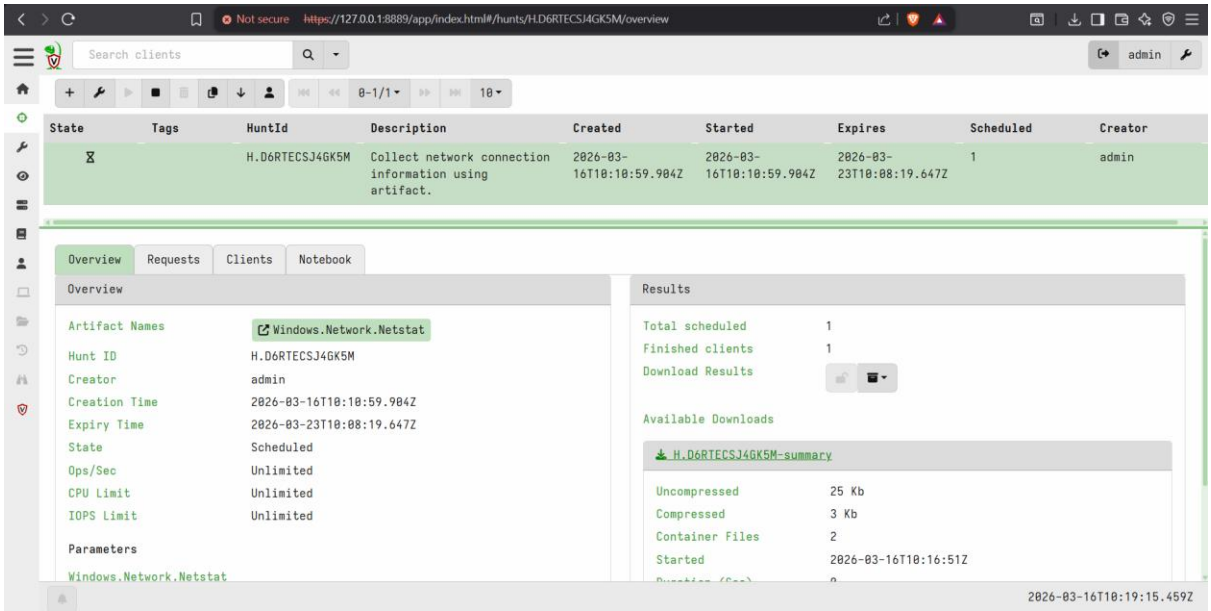
- Evidence files were stored in a secure forensic directory.
- Hash values were recorded immediately after acquisition.
- Original files were preserved in read-only format.
- Copies were used for further analysis.

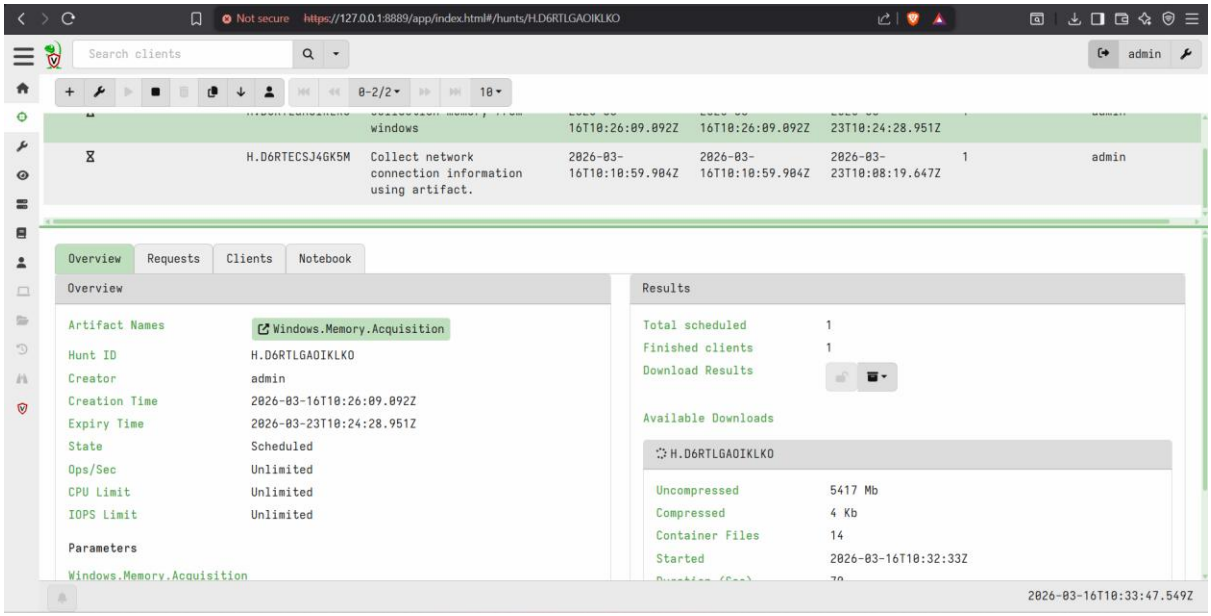
These procedures ensure that forensic artifacts remain reliable for investigation.

Conclusion

This exercise demonstrated the process of collecting and preserving digital forensic evidence in a SOC environment. Using Velociraptor, volatile data such as network connections and memory artifacts were successfully collected from a Windows endpoint. The collected memory evidence was verified using SHA256 hashing to maintain integrity. Proper chain-of-custody documentation was created to ensure accountability and reliability of the collected evidence.

The task highlights the importance of forensic evidence preservation in incident response investigations.






```
C:\Users\USER\Downloads\memory_dump\results>certutil -hashfile "All Windows.Memory.Acquisition.csv" SHA256
SHA256 hash of All Windows.Memory.Acquisition.csv:
8d0ab37cb71ee6121385ca54a9d2e3d53b23afa4b95816a8da154d481d79c92f
CertUtil: -hashfile command completed successfully.

C:\Users\USER\Downloads\memory_dump\results>
```